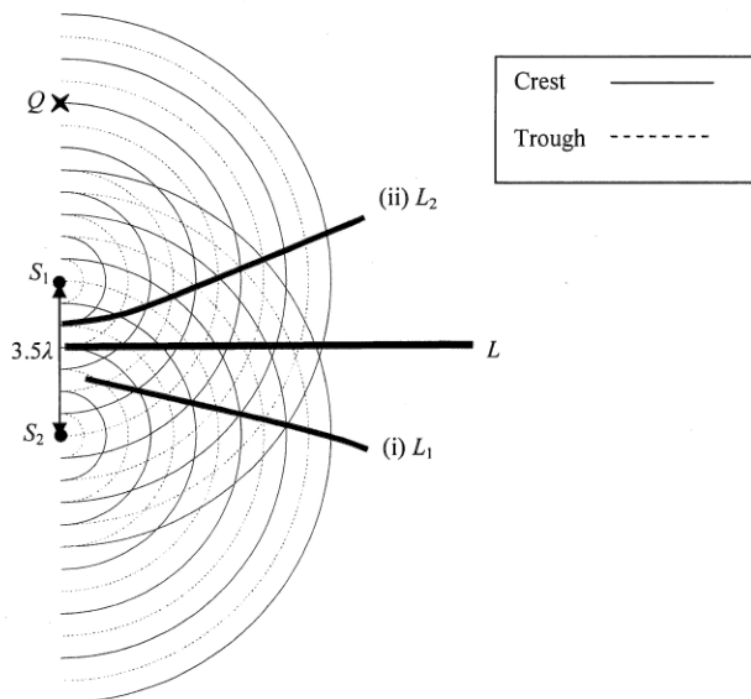


6. (a)



2A

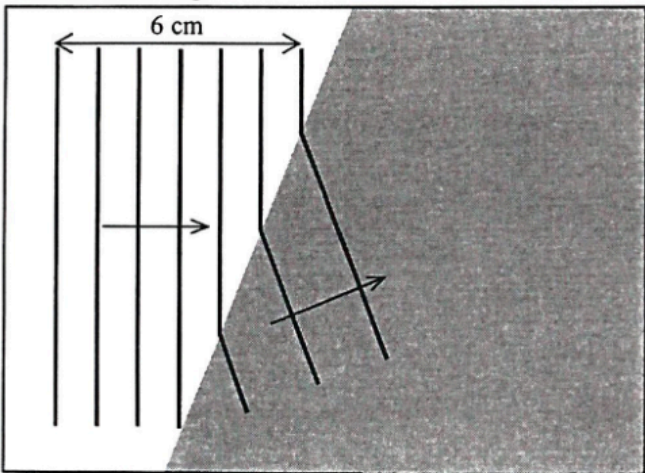
L_1 / L_2 are further from L or separation of $L / L_1 / L_2$ is greater or angle between $L / L_1 / L_2$ larger.

1A

3

7. (a)	(i)	$\tan \theta = 0.38$	1A	Accept 20.8° to 21°
		$\theta = 20.8^\circ$	1	
	(ii)	$d \sin \theta = n\lambda$		Accept $5.90 \times 10^{-7} \text{ m}$ to $5.97 \times 10^{-7} \text{ m}$
		As $d = (\frac{1}{300} \times 10^{-3})$,	1M	
		$(\frac{1}{300} \times 10^{-3}) \times \sin 20.8^\circ = 2\lambda$	1M	
		$\lambda = 5.92 \times 10^{-7} \text{ m (or 592 nm)}$	1A	
			3	
	(iii)	Smaller percentage error in x / the diffraction angle θ .	1A	
		Or Larger x , percentage error decreases.	1A	
			1	
	(b)	Repeat the procedures with the pin on the left-hand side / the other side of the observer.	1A	
		Take the average value of x obtained from both sides to calculate λ .	1A	
			2	

Solution		Marks	Remarks
7. (a)	(i) Increase the separation between the double slit and the screen, D .	1A	
		1	
	(ii) The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
		1	
	(iii) The angular position of the 2 nd order bright dot $\theta = \tan^{-1}\left(\frac{1.56/2}{1.40}\right) = 29.124053^\circ$	1M	
	Grating spacing $d = \frac{10^{-3}}{400} = 2.5 \times 10^{-6} \text{ m}$	1M	
	Applying $d \sin \theta = n\lambda$, Wavelength $\lambda = \frac{2.5 \times 10^{-6} \times \sin 29.12^\circ}{2}$ $= 6.08378 \times 10^{-7} \text{ m}$ $\approx 6.08 \times 10^{-7} \text{ m} (= 608 \text{ nm})$	1A	
		3	
	(b) (i) The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength $\ll$ separation of the two sources), <u>OR</u> λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources $\ll$ separation of the sources and detector), <u>OR</u> a is much smaller than D	1A	
	Any ONE		
	Or Using the fringe separation equation to find the wavelength of sound is not accurate for - λ is comparable to/ NOT much smaller than a - a is NOT much smaller than D	1A	Note : the order of magnitude of the wavelength of sound is about 10^{-1} m
	Any ONE		
		1	
	(ii) For the 1 st order maximum, wavelength $\lambda = \text{path difference } PB - PA$ $= \sqrt{(1+0.5)^2 + 2^2} - \sqrt{(1-0.5)^2 + 2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438 \text{ m}$	1M	
	speed of sound: $v = f\lambda = 750 \times 0.4384$ $= 328.835 \text{ m s}^{-1}$ $\approx 329 \text{ m s}^{-1}$	1M	
		1A	
		3	
			Accept $\lambda = (6.06 \text{ to } 6.10) \times 10^{-7} \text{ m}$
			Accept $v = 328 \text{ m s}^{-1} \text{ to } 330 \text{ m s}^{-1}$

5. (a) (i) wavelength $\lambda = \frac{0.06}{7-1}$ $= 0.01 \text{ m } (= 1 \text{ cm})$	1A 1	
(ii) speed $v = f\lambda = 10 \times 0.01$ $= 0.1 \text{ m s}^{-1} (= 10 \text{ cm s}^{-1})$	1M/1A 1	
(b) (i) frequency = 10 Hz	1A 1	
(ii) 	1A 1A 2	
(iii) Refraction. It is due to the change in wavelengths / wave speeds in different media / depths.	1A 1A 2	

6. (a) Apply $d \sin \theta = m\lambda$, where θ is the deviation angle of the diffracted beam, m is the order of diffraction maximum.		
$\tan \theta = \frac{x/2}{L} \Rightarrow \theta = 24.702430^\circ$ or $\sin \theta = \frac{x/2}{\sqrt{L^2 + (x/2)^2}} = 0.417906$	1M	
$d = \frac{\lambda}{\sin \theta} = \frac{650 \times 10^{-9}}{0.418}$	1M	
$= 1.555375 \times 10^{-6} \text{ m} \approx 1.56 \mu\text{m}$	1A	Accept: $1.56 \mu\text{m} \sim 1.6 \mu\text{m}$
	3	
(b) x will decrease as λ decreases, $\sin \theta$ and θ will decrease.	1A 1A	
	2	